

Volume I.

§ 40. Continuity of the roots.

We close this section with the proof of the theorem:

The roots of an algebraic equation are continuous functions of the coefficients.

We have first to explain the meaning of this theorem.

Let

$$f(x) = x^n + a_1x^{n-1} + a_2x^{n-2} + \dots \quad (1)$$

be an integral rational function of x of degree n . By what has been proved in the preceding paragraphs, $f(x)$ can be decomposed into n linear factors, some of which may be equal to one another. Combining equal factors and denoting the roots by $\alpha, \beta, \gamma, \dots$, we put

$$f(x) = (x - \alpha)^a (x - \beta)^b (x - \gamma)^c \dots, \quad (2)$$

where $a, b, c, \dots$ are positive whole numbers whose sum is equal to n .

The roots $\alpha, \beta, \gamma, \dots$ will change with the coefficients $a_1, a_2, a_3, \dots$; the degree of their multiplicity can also become another one.

We denote the changes of $a_1, a_2, \dots$ by $\varepsilon_1, \varepsilon_2, \dots$ and put

$$\varphi(x) = \varepsilon_1x^{n-1} + \varepsilon_2x^{n-2} + \dots, \quad (3)$$

$$f(x) + \varphi(x) = f_1(x). \quad (4)$$

We surround the points $\alpha, \beta, \gamma, \dots$ by regions of arbitrary smallness, but in such a way that these regions exclude one another; for instance, we enclose the points $\alpha, \beta, \gamma, \dots$ by circular peripheries with radii $\rho, \rho', \rho'', \dots$, and denote these regions by $(\rho), (\rho'), (\rho''), \dots$

If the absolute values of $\varepsilon_1, \varepsilon_2, \dots$ lie below sufficiently small values, then we can prove first of the function $f_1(x)$

that it has no roots outside the regions $(\rho), (\rho'), (\rho''), \dots$,

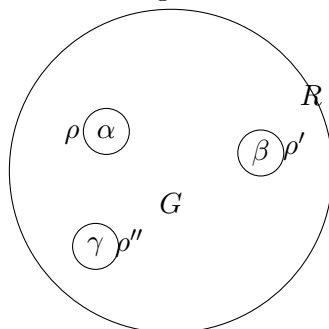
and secondly,

that the number of roots of $f_1(x)$ within (ρ) is exactly a , within (ρ') exactly b , within (ρ'') exactly c , and so on.

In the last part of the theorem, however, it must be observed that, if $f_1(x)$ has multiple roots, these must be counted according to their multiplicity.

We construct, according to §31, IV, in the x -plane a circle with radius R and with the origin as centre, which encloses the regions $(\rho), (\rho'), (\rho''), \dots$, so that outside this circle no more roots of $f_1(x)$ lie, and we denote by G the region lying within this circle, but outside $(\rho), (\rho'), (\rho''), \dots$. It must also be remarked that R can be assumed independently of $\varepsilon_1, \varepsilon_2, \dots$, so long as a definite upper bound is fixed for the absolute values of these quantities.

Fig. 4.



If now x is a point of the region G , then the absolute value of $x - \alpha$ is greater than ρ , and consequently, by (2),

$$|f(x)| > \rho^a \rho'^b \rho''^c \dots \quad (5)$$

If now α_1 is a root of $f_1(x)$, then from (4) it follows that

$$f(\alpha_1) = -\varphi(\alpha_1), \quad (6)$$

and α_1 therefore cannot lie in the region G if one takes care that everywhere within G

$$|\varphi(x)| < \rho^a \rho'^b \rho''^c \dots, \quad (7)$$

which is always possible by sufficient diminution of the upper bound of $\varepsilon_1, \varepsilon_2, \dots$. Thus the first part of our assertion is proved, namely that, when the inequality (7) is satisfied, no roots of $f_1(x)$ lie outside the regions $(\rho), (\rho'), (\rho''), \dots$

To prove the second part, we put

$$f(x) = \psi(x)(x - \alpha)^a, \quad (8)$$

$$\psi(x) = (x - \beta)^b (x - \gamma)^c \dots \quad (9)$$

We now choose a positive number A , which shall however be independent of ρ , in such a way that, as long as x lies in the interior or on the periphery of the region (ρ) ,

$$|\psi(x)| > A \quad (\text{for } |x - \alpha| \leq \rho) \quad (10)$$

remains true. Such a number is obtained, for example, if l is taken equal to half the least of the distances $(\alpha, \beta), (\alpha, \gamma), \dots$ and one puts

$$A = l^{b+c+\dots};$$

then the requirement expressed in (10) is fulfilled at least as long as ρ is smaller than l . If only one point α is present, then $\psi(x) = 1$ is to be put, and for A any proper fraction can be chosen.

Let now $\alpha_1, \alpha_2, \dots$ be the roots of $f_1(x)$ within (ρ) , and let $a_1, a_2, \dots$ be the degrees of their multiplicity; and let $\beta_1, \beta_2, \dots, b_1, b_2, \dots, \gamma_1, \gamma_2, \dots, c_1, c_2, \dots$ have the same meaning for the regions $(\rho'), (\rho''), \dots$. Then

$$n = a_1 + a_2 + \dots + b_1 + b_2 + \dots + c_1 + c_2 + \dots, \quad (11)$$

since the total number of all roots must be equal to n . The function $f_1(x)$ can then be represented as

$$f_1(x) = \psi_1(x)(x - \alpha_1)^{a_1}(x - \alpha_2)^{a_2} \dots, \quad (12)$$

where

$$\psi_1(x) = (x - \beta_1)^{b_1}(x - \beta_2)^{b_2} \dots (x - \gamma_1)^{c_1}(x - \gamma_2)^{c_2} \dots \quad (13)$$

Now we can determine a positive number B independent of $\rho, \rho', \rho'', \dots$ such that, as long as x remains within or on the boundary of (ρ) ,

$$|\psi_1(x)| < B \quad (\text{for } |x - \alpha| \leq \rho). \quad (14)$$

For example, one can choose a magnitude L which is greater than twice the distance of the point α from one of the points $\beta, \gamma, \dots$ and in every case greater than 1, and then assume

$$B > L^n.$$

Now from (4) it follows that

$$|f(x)| < |f_1(x)| + |\varphi(x)|, \quad (15)$$

hence by (8) and (12)

$$|\psi(x)| |x - \alpha|^a < |\psi_1(x)| |x - \alpha_1|^{a_1} |x - \alpha_2|^{a_2} \cdots + |\varphi(x)|. \quad (16)$$

If we now assume x on the periphery of ρ , and therefore put

$$|x - \alpha| = \rho,$$

then

$$|x - \alpha_1| < 2\rho, \text{quad} |x - \alpha_2| < 2\rho, \dots,$$

and consequently, by (10) and (16),

$$\rho^a A < (2\rho)^{a_1+a_2+\cdots} B + |\varphi(x)|. \quad (17)$$

We now choose the upper bounds for the coefficients $\varepsilon_1, \varepsilon_2, \dots$ of φ so small that

$$|\varphi(x)| < \rho^a A'$$

where A' denotes a number which is smaller than A ; then from (17) it follows that

$$A - A' < 2^a \rho^{-a+a_1+a_2+\cdots} B. \quad (18)$$

But, for sufficiently small ρ , this would no longer be possible if a were greater than the sum $a_1 + a_2 + \cdots$. Hence

$$a \leq a_1 + a_2 + \cdots. \quad (19)$$

In the same way one proves that

$$b \leq b_1 + b_2 + \cdots, \quad c \leq c_1 + c_2 + \cdots, \quad \dots \quad (20)$$

But since the sums of the left sides as well as of the right sides in the inequalities (19), (20) must be equal to n , only equality signs can hold, and therefore

$$a = a_1 + a_2 + \cdots, \quad b = b_1 + b_2 + \cdots, \quad c = c_1 + c_2 + \cdots, \quad \dots,$$

whereby the second part of our theorem is also proved.

We shall add to the theorem just proved the following remark concerning the case of multiple roots.

The necessary and sufficient condition that $f(x) = 0$ should have multiple roots at all is that $f(x)$ and $f'(x)$ should have a common factor. Therefore, if the algorithm of the greatest common divisor is applied to these two functions according to the principles developed in the first section, the condition for common factors is obtained by setting the last constant remainder equal to zero, in the form of an equation between the coefficients,

$$F(a_1, a_2, \dots, a_n) = 0,$$

where F is an integral rational function of the arguments $a_1, a_2, \dots, a_n$, which is called the discriminant of f , and whose properties and mode of formation will be treated more fully in the next section. In any case, since equations of degree n without multiple roots do exist at all, F cannot vanish identically.

Consider now

$$F(a_1 + \varepsilon_1, a_2 + \varepsilon_2, \dots, a_n + \varepsilon_n).$$

This function too, if $a_1, a_2, \dots, a_n$ are definite and $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n$ are indeterminate numbers, will not vanish identically; and one can dispose of the ε 's so that its value comes out different from zero, even when any arbitrary upper bound has been fixed for the absolute values of the ε 's.¹

¹This will only later be rigorously justified. For the time being we presuppose it in these preliminary remarks, on which no further conclusions will be founded.

One therefore sees from these considerations how an a -fold root α of $f(x)$, under continuous variation of the coefficients, radiates apart into a simple roots, so that from this point of view too one is justified in considering the a -fold root as having arisen from the coincidence of a simple roots.

If we do not assume the coefficient of the highest power of x equal to 1, and therefore examine the equation in the form

$$a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n = 0, \quad (21)$$

then, if $a_n = 0$, one or more of the roots are equal to zero, and equation (21) can be divided by x or by a higher power of x , whereby an equation of correspondingly lower degree arises in which the last term is different from zero. We shall therefore assume from the outset that a_n is different from zero, and now put in (21)

$$x = \frac{1}{y}. \quad (22)$$

By multiplication by y^n and division by a_n , (21) then passes into

$$y^n + \frac{a_{n-1}}{a_n}y^{n-1} + \cdots + \frac{a_1}{a_n}y + \frac{a_0}{a_n} = 0, \quad (23)$$

and we can apply to the roots of this equation the theorem just stated, by assuming $a_0 = 0$, so that one of the roots of (23) vanishes. We thus obtain the theorem:

One can take a_0 in the equation (21) so small that one of its roots becomes large beyond all bounds, while the others differ arbitrarily little from the roots of the equation of degree $n - 1$

$$a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n = 0.$$

This is also expressed by saying that, as a_0 vanishes, one of the roots of (21) becomes infinite.

If a_1 also vanishes, or a_1 and a_2 , and so on, two or three roots become infinite.

This justifies the expression sometimes used, that an equation of degree n passes, by the becoming-infinite of one root, into an equation of degree $n - 1$.